NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-94/2C
26 February 1971
SUPERSEDING
MIL-R-94/2B
5 July 1966

PERFORMANCE SPECIFICATION

RESISTORS, VARIABLE, COMPOSITION, STYLE RV5

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RV5 composition, variable, resistors 1/.

2. APPLICABLE DOCUMENT

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-94 - Resistors, Variable, Composition, General Specification for.

(Copies of specifications, standards, drawings, and publications required by supplies in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-94, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.2.1 Mounting and locking nuts. The mounting nut shall be 3/32-inch thick and shall measure 3/8 inch across the hexagonal flats. The locking nut shall be 1/8-inch thick and shall measure 3/8 inch across the hexagonal flats. Thread size shall be 1/4-32NEF-2B.

3.2.2 Internal-tooth lockwasher. Internal-tooth lockwasher of suitable size shall be applied.

3.2.3 Retainer rings. If retainer rings are used, they shall be not thicker than 0.032 inch.

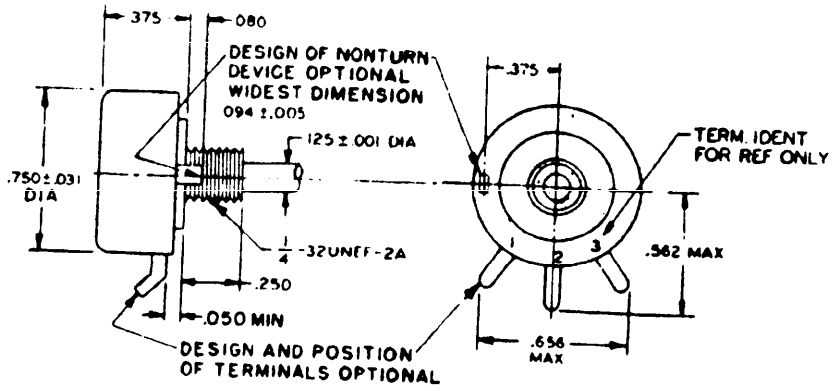
3.3 Total mechanical rotation. The total mechanical rotation without switch shall be within the limits of 295° and 305°.

3.4 Shaft length. The length of operating shall be in accordance with table I.

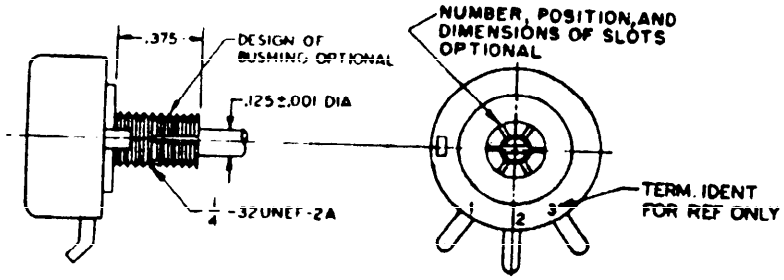
1/ Not to be used for new design. For replacement purposes only. For new design use style RV6.

FSC 5905

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:54Z
Check the source to verify that this is the current version before use.

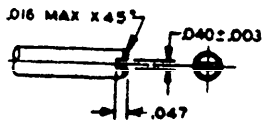


STANDARD-BUSHING TYPE



LOCKING-BUSHING TYPE

INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.047	1.19	.250	6.35
.003	.08	.050	1.27	.375	9.53
.005	.13	.080	2.03	.562	14.27
.016	.41	.094	2.39	.656	16.66
.031	.79	.125	3.18	.750	19.05
.040	1.02				



SLOTTED SHAFT

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is .xxx±.016(.41 mm).

FIGURE 1. Style RV5 resistors.

Source: https://assist.dla.mil -- Downloaded: 202610-16T04:54Z
Check the source to verify that this is the current version before use.

TABLE I. Length of operating shaft.

Symbol	Length of shaft from mounting surface of resistors. Nominal	
	Bushing N	Bushing L
	inches (±1/32)	inches (±1/32)
B		1/2
D	7/8	7/8 1/
L	3/8	

1/ For replacement purposes only, not for new design.

3.5 Nominal total resistance and rated continuous working voltages. The nominal total resistance and rated continuous working voltage (RCWV) shall be as specified in table II.

TABLE II. Nominal total resistance and rated continuous working voltage.

Nominal total resistance	RCWV 1/ (volts)		Nominal total resistance	RCWV 1/ (volts)	
	Taper A	Tapers C and F		Taper A	Tapers C and F
Ohms			Megohms		
250	11	---	.10	224	160
500	16	---	.20 2/	316	200
1,000	22	16	.25	350	200
2,000 2/	31	22	.50	350	200
2,500	35	25	1.0	350	200
5,000	50	36	2.0	350	200
10,000	71	50	2.5	350	200
20,000 2/	100	70			
25,000	112	80			
50,000	158	112			

1/ Rated continuous working voltage at 70°C.
2/ For replacement purposes only. Not for new design.

3.6 Power rating. The power rating shall be 1/2 watt for taper-A resistors and 1/4 watt for tapers-C and -F resistors.

3.7 Torque.

3.7.1 Operating. The torque required to effect rotation shall be not less than 0.50 ounce-inch nor more than 6 ounce-inches.

3.7.2 Stop. The torque applied through the operating shaft to the stops shall be 5 pound-inches.

3.7.3 Locking bushing. For the locking-bushing type, the locking nut shall be tightened with a torque of 10 pound-inches. After the locking nut is tightened, the contact arm shall not move when a torque of 20 ounce-inches is applied to the shaft.

Source: https://assist.dla.mil -- Downloaded: 202610-16T04:54Z
Check the source to verify that this is the currentversion before use.

3.8 Thermal cycling. The resistors shall not change in resistance in excess of 6 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-94, and as specified herein.

4.2 Dielectric withstanding voltage. In the dielectric withstanding voltage test, the applied potential at reduced barometric pressure shall be 350 volts, rms.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-94.

6. NOTES

6.1 Notes. The notes specified in MIL-R-94 are applicable to this specification.

6.2 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - MI, MU, EL
Navy - EC, AS, OS
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5905-0745)

* U. S. GOVERNMENT PRINTING OFFICE: 1971-423-894/5148

Source: https://assist.dla.mil -- Downloaded: 202610-16T04:54Z
Check the source to verify that this is the currentversion before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

COMMANDING OFFICER
U.S. ARMY ELECTRONICS COMMAND
ATTN: AMSEL-TD-S
FORT MONMOUTH, N.J. 07703

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:54Z
Check the source to verify that this is the current version before use.

S/N 0102-LF-001-4270 (See 5.10).